

Life Star Academy By Sir Afzal

Cs301 MidTerm Object File:04

(Lectures from 1 to 23)

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Question No: 1 (Marks: 1) - Please choose one

In an array we can store data elements of different types.

- ▶ True
- ▶ **False** (Page 7)

Question No: 2 (Marks: 1) - Please choose one

In an array list the current element is

- ▶ **The first element** [Click here for detail](#)
- ▶ The middle element
- ▶ The last element
- ▶ The element where the current pointer points to

Question No: 3 (Marks: 1) - Please choose one

Which one of the following calling methods does not change the original value of the argument in the calling function?

- ▶ None of the given options
- ▶ **Call by passing the value of the argument** [Click here for detail](#)
- ▶ Call by passing reference of the argument
- ▶ Call by passing the address of the argument

Question No: 4 (Marks: 1) - Please choose one

Which one of the following statements is NOT correct?

- ▶ **Array size can be changed after its creation.** [Click here for detail](#)
- ▶ Link List size can be changed after its creation
- ▶ Binary Search Tree size can be changed after its creation
- ▶ AVL Tree size can be changed after its creation

Question No: 5 (Marks: 1) - Please choose one

Suppose that the class declaration of SomeClass includes the following function prototype. `bool LessThan( SomeClass anotherObject );`

Which of the following tests in the client code correctly compares two class objects alpha and beta?

- ▶ `if (alpha < beta)`
- ▶ **`if (alpha.LessThan(beta))`** [Click here for detail](#)
- ▶ `if (LessThan(alpha, beta))`
- ▶ `if (LessThan(alpha).beta)`

Question No: 6 (Marks: 1) - Please choose one

A queue is a----- data structure, whereas a stack is a----- data structure.

- ▶ **FIFO, LIFO** (Page 161,54)
- ▶ LIFO,FIFO
- ▶ none of these
- ▶ both of these

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Question No: 7 (Marks: 1) - Please choose one

Which one of the following operators has higher priority than all of others?

- ▶ **Multiplication operator** [Click here for detail](#)
- ▶ Minus operator
- ▶ Plus operator
- ▶ Exponentiation operator

Question No: 8 (Marks: 1) - Please choose one

Each node in Binary Search Tree has

- ▶ 1 pointer
- ▶ **2 pointers** [Click here for detail](#)
- ▶ 3 pointers
- ▶ 4 pointers

Question No: 9 (Marks: 1) - Please choose one

Four statements about trees are below. Three of them are correct. Which one is INCORRECT?

- ▶ Trees are recursively defined multi-dimensional data structures tree
- ▶ The order of a tree indicates a maximum number of children allowed at each node of the
- ▶ A search tree is a special type of tree where all values (i.e. keys) are ordered
- ▶ **If Tree1's size is greater than Tree2's size, then the height of Tree1 must also be greater than Tree2's height.** [Click here for detail](#)

Question No: 10 (Marks: 1) - Please choose one

Which of the following is "TRUE" about arrays,

- ▶ We can increase the size of arrays after their creation.
- ▶ We can decrease the size of arrays after their creation.
- ▶ We can increase but can't decrease the size of arrays after their creation.
- ▶ **We can neither increase nor decrease the array size after their creation.** [Click here for detail](#)

Question No: 11 (Marks: 1) - Please choose one

Searching an element in an AVL tree take maximum in AVL tree,

- ▶ $\log_2(n+1)$
time (where n is no. of nodes)
- ▶ $\log_2(n+1) - 1$
- ▶ **$1.44 \log_2 n$** (Page 227)
- ▶ $1.66 \log_2 n$

Question No: 12 (Marks: 1) - Please choose one

There is/are case/s for rotation in an AVL tree,

- ▶ 1
- ▶ 3
- ▶ 2
- ▶ **4** (Page 229)

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Question No: 13 (Marks: 1) - Please choose one

Consider the following statements.

- (i) A binary tree can contain at least 2^L Nodes at level L .
- (ii) A complete binary tree of depth d is a binary tree that contains 2^L Nodes at each level L between 0 and d , both inclusive.
- (iii) The total number of nodes (T_n) in a complete binary tree of depth d is $2^{d+1} - 1$.
- (iv) The height of the complete binary tree can be written as $h = \log_2(T_n + 1) - 1$ where T_n is Total number of Nodes.

Which one of the following is correct in respect of the above statements regarding the Binary trees?

- ▶ (i) and (iii) only
- ▶ (i), (ii) and (iii) only
- ▶ (ii) and (iii) only
- ▶ (ii), (iii) and (iv) only [Click here for detail](#)

Question No: 14 (Marks: 1) - Please choose one

Consider the following infix expression.

$5 + 6/2$

If one converts the above expression into postfix, what would be the resultant expression?

- ▶ $56/ + 2$
- ▶ **$5\ 6\ 2\ /\ +$** (Page 66)
- ▶ $5\ 6\ /\ 2 +$
- ▶ $/62 + 5$

Question No: 15 (Marks: 1) - Please choose one

Which of the following is a non linear data structure?

- ▶ Linked List
- ▶ Stack
- ▶ Queue
- ▶ **Tree** (Page 112)

Question No: 16 (Marks: 1) - Please choose one

“+” is a *operator*.

- ▶ Unary
- ▶ **Binary** (Page 64)
- ▶ Ternary
- ▶ None of the above